

Figure 1.11 Mercerizing Machine

Process

The mercerizing process involves these three steps:

Step 1: Impregnation of the material in relaxed state with cold caustic solution of required strength and wet-ability.

Step 2: Stretching while the material is still impregnated in the caustic solution.

Step 3: Washing off the caustic soda from the material while keeping the material still in the stretched state.

An optional last step in the process is passing the thread over an open flame; this incinerates stray fibres, improving the fabric's appearance. This is known as "gassing the thread" due to the gas burner that is typically used.

Mercerizing can take place directly on grey cloth, or can also be done after bleaching. It can be done with or without tension in both cold and hot conditions. In both cases the mercerised cotton has an increased affinity for both reactive and direct cotton dyes, water and an increased strength.

Advantages of Mercerization

- Larger dyeing affinity
- Larger dimensional stability of the articles
- Increasing of the lustre
- Increasing of the tensile strength
- Better covering of dead cotton
- Improving touch

1.3.12 Heat Setting

Heat setting is an important part in textile finishing. It is one of the functional finishes which are carried out mainly on synthetic fabrics. It eliminates the internal tensions generated during the manufacture of fibre and fixes the new state by rapidly cooling it. Heat setting fixes the fabrics in the relaxed state and thus avoids subsequent shrinkage or creasing of fabric. Presetting of goods make it possible to use higher temperature for setting without considering the sublimation properties of dyes and also has a favourable effect on dyeing behaviour and running properties of goods. On the other hand, post-setting can be combined with some other operations such as thermasol dyeing or optical brightening of polyester. Post-setting as a final finish is useful to achieve high dimensional stability, along with desired handle.



Figure 1.12 Heat Setting Machine

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